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| **Course Title** | Digital Logic Design | **Course Code** | EE1005 |
| **Department** | Department of Electrical Engineering (DEE) | **Campus** | Lahore |
| **Knowledge Profile** | Engineering Design (WK5) | **Credit Hrs.** | 3+1 |
| **Knowledge Area** | Computer Engineering (KA01) | **Grading Scheme** | Relative |
| **HEC Knowledge Area** | Engineering Foundation | **Applicable From** | Spring 2023 |
| **Pre-requisite(s)** | | | |

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| **Course Objective** | To introduce the classical logic design techniques for analysis and design of combinational and sequential circuits. |

| No. | Assigned Program Learning Outcome (PLO) |
|---|---|
| 1 | An ability to apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems. |
| 3 | An ability to design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations. |

**I = Introduction, R = Reinforcement, E = Evaluation, A = Assignment, Q = Quiz, M = Midterm, F=Final, L = Lab, P = Project, W = Written Report.**

| No. | Course Learning Outcome (CLO) Statements | Assessment Tools | Taxonomy Levels | PLO |
|---|---|---|---|---|
| 1 | Describe various number systems and perform arithmetic operations and base conversions. | M1, A1 | C2 | 1 |
| 2 | Apply Boolean Algebra and K-map methods to optimize logic circuits | Q1, M1, A2 | C3 | 1 |
| 3 | Design combinational circuits using functional blocks | Q2, M2, F | C5 | 3 |
| 4 | Analyze elements of sequential circuits and design finite state machines | M2, A3 | C4 | 1 |
| 5 | Design of sequential circuits for various applications | Q3, F | C4 | 3 |

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| **Text Books** | **Title** | Logic and Computer Design Fundamentals (5th Edition, 2015) |
| | **Author** | M. Morris Mano & Charles R. Kime |
| | **Publisher** | Prentice Hall |
| **Reference Books** | **Title** | |
| | **Author** | |
| | **Publisher** | |

| Week | Course Contents/Topics | Chapter* | CLO* |
|---|---|---|---|
| 01 | DIGITAL COMPUTERS AND INFORMATION, Digital computers and Binary Numbers, Other base numbers (base-8, base-16 etc.), Number base conversions. | 1 | 1 |
| 02-03 | COMBINATIONAL LOGIC CIRCUITS, Binary Logic and Logic Gates, Axioms and Basic Theorems of Boolean Algebra, Boolean Functions and their implementation, Canonical and Standard Forms (Minterms, Maxterms, Conversions), Minimization of Boolean functions using basic theorems and K-Maps, Don't Care States, Universal gates and implementation of Boolean functions using universal gates | 2 | 2 |
| 04-05 | COMBINATIONAL LOGIC DESIGN, Combinational Circuits, Decoders / Encoders, Multiplexers / Demultiplexer, Binary Adders (Half Adder, Full Adders, Binary Ripple Carry Adder) | 3 | 3 |
| 05-07 | Binary Subtractor, Binary Adder/Subtractor, Code Conversion, Magnitude Comparator. | 3 | 3 |
| 08-12 | SEQUENTIAL CIRCUITS, Sequential Circuits (FSM), Latches, Flip Flops, Type of Flip Flops, Analysis of Sequential Circuits, Design Procedures, State diagrams and state tables. | 4 | 4 |
| 13-14 | REGISTERS AND COUNTERS, Registers, Counters, Synchronous/Asynchronous, Shift Registers, Serial Shift Registers, Loading Registers, Parallel Registers, Ripple Counters, Synchronous Binary and other Counters. | 7 | 5 |
| 15-16 | MEMORY AND PROGRAMMABLE LOGIC DEVICES, ROM (Read-Only Memories), PLA (Programmable Logic Array) Devices, RAM (Random Access Memory), Static and Dynamic RAM, Array of RAM ICs, Memory construction using RAM Integrated Circuits, Basic Architecture of FPGAs | 6, Material will be provided | 5 |

***Reference book chapters are given in brackets**

| Assessment Tools | Weightage |
|---|---|
| **Quizzes, Assignments** | 20% |
| **Midterm (I+II)** | 15% + 15% |
| **Final Exam** | 50% |